

**FBISE CLASS 9 MATHEMATICS****Chapter 1: Real Numbers****Practice Test - MCQs and Short Questions**

Based on the provided Chapter 1 question-bank file

**Name:** \_\_\_\_\_ **Roll No.:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Instructions:** This practice test contains two sections only. Section A has 20 multiple-choice questions and Section B has 8 short questions. Attempt all questions for comprehensive practice. Use laws of indices/rational exponents and scientific notation as required. Assume variables are positive real numbers wherever a rational exponent or radical requires it.

**SECTION A - MULTIPLE CHOICE QUESTIONS***Question 1: Encircle the correct option. (20 MCQs)*

1.  $\left(\frac{1}{9}\right)^{-3/2}$  is equal to

(a) 27

(c)  $\pm 27$ (b)  $\frac{1}{27}$ (d)  $\pm \frac{1}{27}$ 

2. The scientific notation of 0.00567 is

(a)  $5.67 \times 10^5$ (c)  $5.67 \times 10^{-2}$ (b)  $5.67 \times 10^{-3}$ (d)  $5.67 \times 10^2$ 

3.  $(x^{2/3})^9$  is equal to

(a)  $x^8$ (c)  $x^{4/9}$ (b)  $x^6$ (d)  $x^{3/2}$ 

4.  $5.67 \times 10^4$  in ordinary form is

(a) 0.000567

(c) 567000

(b) 0.567

(d) 56700

5.  $\left(\frac{1}{4}\right)^{-1/2}$  is equal to

(a)  $\frac{1}{2}$ (c)  $\sqrt{2}$ 

(b) 2

(d) 16

6. The scientific notation of 345.6 is

(a)  $3.456 \times 10^2$ (c)  $3.456 \times 10^{-2}$ (b)  $3.456 \times 10^3$ (d)  $3.456 \times 10^{-3}$ 

7. The radical form of  $x^{-3/2}$  is

(a)  $\sqrt[3]{x^2}$ (c)  $\sqrt{x^3}$ (b)  $\frac{1}{\sqrt{x^3}}$ (d)  $\frac{1}{\sqrt[3]{x^2}}$ 

8. Which option is the scientific notation of  $0.25^2$ ?

(a)  $625 \times 10^{-4}$ (c)  $6.25 \times 10^{-2}$ (b)  $62.5 \times 10^{-3}$ (d)  $0.625 \times 10^{-1}$ 

9.  $\sqrt[3]{a^5}$  is equal to

(a)  $a^{3/5}$ (c)  $a^{5/3}$ (b)  $a^{5/2}$ (d)  $a^{3/2}$

## SECTION A - MULTIPLE CHOICE QUESTIONS (continued)

10.  $\left(\frac{1}{16}\right)^{-3/2}$  is equal to

- (a)  $\frac{1}{4}$   
(c) 16

- (b) 4  
(d) 64

11. The scientific notation of 4500 is

- (a)  $4.5 \times 10^3$   
(c)  $4.5 \times 10^2$

- (b)  $45 \times 10^3$   
(d)  $0.45 \times 10^5$

12.  $\left(\frac{1}{9}\right)^{-1/2}$  is equal to

- (a) 3  
(c) 9

- (b) 6  
(d) 81

13. The scientific notation of 71000 is

- (a)  $7.1 \times 10^2$   
(c)  $7.1 \times 10^3$

- (b)  $0.71 \times 10^2$   
(d)  $7.1 \times 10^4$

14.  $(27x^{-1})^{-2/3}$  simplifies to

- (a)  $\frac{\sqrt[3]{x^2}}{9}$   
(c)  $\frac{\sqrt[3]{x^2}}{8}$

- (b)  $\frac{\sqrt{x^3}}{9}$   
(d)  $\frac{\sqrt{x^3}}{8}$

15.  $\sqrt[7]{x}$  in exponential form is

- (a)  $x$   
(c)  $x^{1/7}$

- (b)  $x^7$   
(d)  $x^{7/2}$

16.  $4^{2/3}$  written with a radical sign is

- (a)  $\sqrt[3]{4^2}$   
(c)  $\sqrt{4^2}$

- (b)  $\sqrt[3]{4^3}$   
(d)  $\sqrt{4^6}$

17. In  $\sqrt[3]{35}$ , the radicand is

- (a) 3  
(c) 35

- (b)  $\frac{1}{3}$   
(d) none of these

18.  $\left(\frac{25}{16}\right)^{-1/2}$  is equal to

- (a)  $\frac{5}{4}$   
(c)  $-\frac{5}{4}$

- (b)  $\frac{4}{5}$   
(d)  $-\frac{4}{5}$

19.  $\left(\frac{8}{125}\right)^{-4/3}$  is equal to

- (a)  $\frac{625}{16}$   
(c)  $\frac{125}{8}$

- (b)  $\frac{16}{625}$   
(d)  $\frac{8}{125}$

20.  $\frac{4(3)^n}{3^{n+1} - 3^n}$  simplifies to

- (a) 1  
(c) 3

- (b) 2  
(d) 4

**SECTION B - SHORT QUESTIONS**

Questions 2-9: Simplify/show the required result using appropriate laws of indices. Show essential working.

**Q2.** Simplify

$$\left[ \left( x^2 y^{-3} z^4 \right)^5 \times \left( x^{-4} y^7 z^{-2} \right)^3 \right]^{1/2}.$$

**Q3.** Simplify

$$\left[ \left( 2a^{1/2} b^3 \right)^4 \times \left( 4a^{-2} b^{-6} \right)^{-1/2} \right]^{1/3}.$$

**Q4.** Simplify

$$\left[ (125)^{1/3} (25)^{1/2} + (64)^{2/3} \times 6 + (8)^{2/3} \right]^{-2/3}.$$

**Q5.** Simplify

$$\frac{(2x^2)^{-4}}{(-4y^{-1}z^2)^{-2}}.$$

## SECTION B - SHORT QUESTIONS (continued)

**Q6.** Simplify

$$\sqrt{\frac{(216)^{2/3}(25)^{1/2}}{(0.04)^{-3/2}}}.$$

**Q7.** Simplify

$$\frac{2^{1/3}(27)^{1/3}(60)^{1/2}}{(180)^{1/2}(4)^{-1/3}(9)^{1/4}}.$$

**Q8.** Show that

$$\left(\frac{x^a}{x^b}\right)^{a+b} \left(\frac{x^b}{x^c}\right)^{b+c} \left(\frac{x^c}{x^a}\right)^{c+a} = 1.$$

**Q9.** Simplify

$$\left(\frac{x^{-2}y^{-1}z^{-4}}{x^4y^{-3}z^0}\right)^{-3}.$$

*End of Test*